

Maps

This is a JupyterLab Notebook summarizing the code I wrote to make some maps of rivers using Python and Geopandas. Eventually, I found some data for Lewis and Clark's expedition (Corps of Discovery).

But before we get to that, here is an outline of the general situation.

Background

On my machines, running macOS, I have Python 3 installed via Homebrew.

It is in the default locations. For the mini, that is `/usr/local/bin`, while on my new Macbook it is in `/opt/homebrew/bin`.

It seems that these python installations are only used with a virtual environment. To set one up, just `cd` into a convenient directory and do

```
/opt/homebrew/bin/python3 -m venv maps
```

That makes a virtual environment with all its supporting files stored in the `maps` directory under whatever directory you were in when you invoked this command.

I like working on the desktop so from `~/Desktop` do

```
source ~/Programming/maps/bin/activate
```

```
(maps) >
```

The prompt will change to `(maps) >`.

That's quite a long command. It's easy to make an alias in `~/.zshrc` such as `alias activate="source ~/Programming/maps/bin/activate"`.

Now that the venv is active you can do:

```
(maps) > pip install --upgrade pip
```

```
(maps) > python -m pip install matplotlib
```

```
(maps) > python -m pip install geopandas
```

```
(maps) > python -m pip install jupyterlab
```

The correct version of python (the one used to make the venv) can be called simply as `python`. Invoking `pip` as a module works better for me than calling it directly.

Geopandas

Pandas is a popular Python module for working with dataframes.

A dataframe is a table with possibly dissimilar columns. For example, one might have rows of individual amino acids, and columns with various things like molecular weight, hydrophobicity, and so on.

A GeoDataFrame (I often call the variable `gdf`) is a dataframe that contains, among other things, coordinates for geographical objects like polygons and line strings, which can represent state boundaries and rivers. Geopandas is a popular Python module for working with this kind of data.

```
In [1... import matplotlib.pyplot as plt
import geopandas as gpd
```

We start with a geodataframe containing points along the boundaries of selected US states in the Pacific Northwest.

```
In [2... dbpath = '/Users/telliott/data/'
fn = 'OR_WA_ID_MT_WY.shp.zip'
nw_states = gpd.read_file(dbpath + fn)
```

Various methods exist to access the individual rows. The 'NAME' column contains the names of states. The 'STATE' column contains the FIPS code for each.

```
In [3... nw_states.columns
```

```
Out[3... Index(['GEO_ID', 'STATE', 'NAME', 'LSAD', 'CENSUSAREA',  
'geometry'], dtype='object')
```

```
In [4... nw_states['NAME']
```

```
Out[4... 0      Oregon  
1    Washington  
2      Idaho  
3    Montana  
4    Wyoming  
Name: NAME, dtype: object
```

```
In [5... nw_states['STATE']
```

```
Out[5... 0      41  
1      53  
2      16  
3      30  
4      56  
Name: STATE, dtype: object
```

```
In [6... result = nw_states[['STATE', 'NAME']]  
result
```

```
Out[6...   STATE  NAME  
0      41  Oregon  
1      53 Washington  
2      16   Idaho  
3      30  Montana  
4      56  Wyoming
```

The code in the cell above appears to access multiple columns. What it really does in addition is to construct and return a new GeoDataFrame.

The 'geometry' column contains objects representing the state outlines as POLYGON or MULTIPOLYGON objects. Washington is a MULTIPOLYGON because it includes a number of islands.

```
In [7... nw_states['geometry']
```

```
Out[7... 0    POLYGON ((-121.92224 45.64908, -121.90827 45.6...
1    MULTIPOLYGON (((-122.51953 48.28831, -122.5227...
2    POLYGON ((-111.04416 43.02005, -111.04413 43.0...
3    POLYGON ((-105.0384 45.00034, -105.07661 45.00...
4    POLYGON ((-110.04848 40.99755, -110.12164 40.9...
Name: geometry, dtype: geometry
```

This dataframe has already been filtered from a larger one containing all 50 states plus DC and Puerto Rico. Let's load the larger one and show how to obtain what we have above.

For some reason, geopandas doesn't like the zip file. Usually it's OK.

```
In [8... #fn='gz_2010_us_040_00_5m.zip'
fn='gz_2010_us_040_00_5m'
gdf = gpd.read_file(dbpath + fn)
print(gdf.shape)
```

```
(52, 6)
```

So 52 entries as rows, and each with 6 columns of attributes. The states we want to select are: OR, WA, ID, MT, WY. Here is one way to do it.

```
In [9... L = ['Oregon', 'Washington', 'Idaho',
      'Montana', 'Wyoming']
sel = gdf['NAME'].isin(L)
```

```
In [1... sub = gdf[sel]
print(sub.shape)
```

```
(5, 6)
```

The `sel` variable is a series of boolean values. The code is otherwise straightforward except for the use of `isin`, which isn't standard Python but a Pandas thing.

```
In [1... sel.iloc[10:14]
```

```
Out[1... 10    False
         11    False
         12     True
         13    False
         Name: NAME, dtype: bool
```

`gdf[sel]` filters for rows with a value of `True`.

`iloc` is a Pandas way of indexing by a numerical index. It is distinguished from `loc`, which uses labels.

apply

Another way to do the same thing is to supply a named function, or more Pythonically, a `lambda` expression.

```
In [1... sel = gdf['NAME'].apply(lambda r: r in L)
         sub = gdf[sel]
         print(gdf.shape, sub.shape)
```

```
(52, 6) (5, 6)
```

Here's another example of `apply`

```
In [1... def f(e):
         if e[0] in list('OWIM'):
             return True
         return False

         sel = [f(e) for e in gdf['NAME']]
         gdf[sel]['NAME']
```

```

Out[1... 12          Idaho
          13        Illinois
          14        Indiana
          15          Iowa
          19        Maine
          20        Maryland
          21    Massachusetts
          22        Michigan
          23        Minnesota
          24        Mississippi
          25        Missouri
          26        Montana
          35          Ohio
          36        Oklahoma
          37        Oregon
          47        Washington
          48    West Virginia
          49        Wisconsin
          50        Wyoming
Name: NAME, dtype: object

```

And yet a third way is to construct a series of logical expressions.

```

In [1... OR = gdf['NAME'] == 'Oregon'
          WA = gdf['NAME'] == 'Washington'
          ID = gdf['NAME'] == 'Idaho'
          MT = gdf['NAME'] == 'Montana'
          WY = gdf['NAME'] == 'Wyoming'
          sub = gdf[OR | WA | ID | MT | WY]
          print(gdf.shape, sub.shape)

```

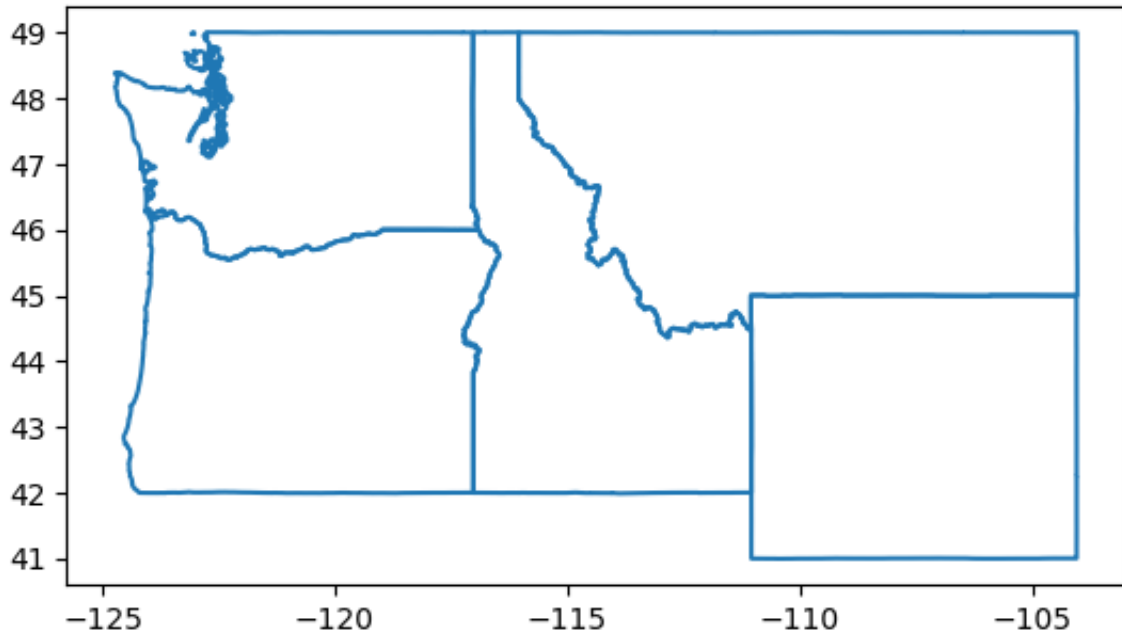
```
(52, 6) (5, 6)
```

Since we have `matplotlib`, let's plot the data. A simple approach is:

```

In [1... nw_states.boundary.plot()
          plt.show()

```



To save the plot to a file

```
In [1... ofn = 'figs/example.png'
plt.savefig(ofn,dpi=300)
```

<Figure size 640x480 with 0 Axes>

As written, this code saves a blank `png` file. The reason is that `plt.show()` was called above, and you can't do both.

It would be nice to have some labels. Let's add a column to the data with the two letter abbreviations.

```
In [1... L = ['OR', 'WA', 'ID', 'MT', 'WY']
nw_states = nw_states.assign(abbrev = L)
nw_states['abbrev']
```

```
Out[1... 0    OR
1    WA
2    ID
3    MT
4    WY
Name: abbrev, dtype: object
```

```
In [1... def f(e):
    return e.representative_point().coords[:,0]
```

The `[0]` at the end is because the coordinates are like `[(x,y)]`.

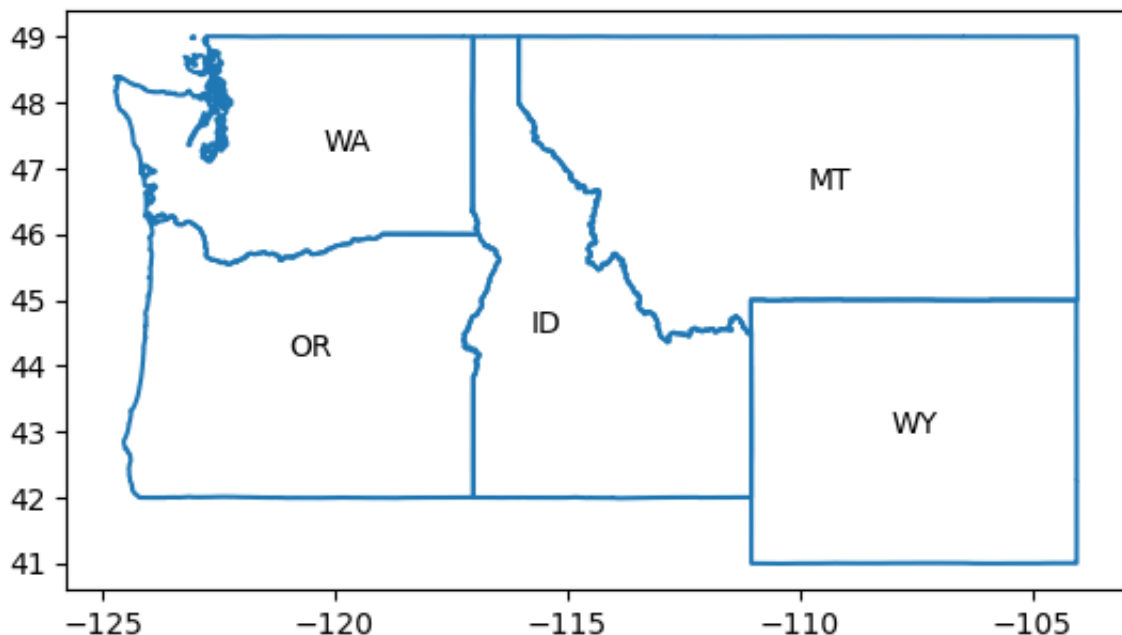
```
In [1... nw_states['coords'] = nw_states['geometry'].apply(f)
nw_states['coords']
```

```
Out[1... 0    (-120.51490187597771, 44.139333893360046)
1    (-119.73301862372591, 47.2734280475521)
2    (-115.46324561457074, 45.494265999999996)
3    (-109.34110685676336, 46.681627999999996)
4    (-107.54854862632588, 42.987824)
Name: coords, dtype: object
```

One can use `iterrows` to iterate through the data. We adjust the position of the label for Idaho.

```
In [2... nw_states.boundary.plot()
for i,row in nw_states.iterrows():
    x,y = row['coords']
    if row['NAME'] == 'Idaho':
        y -= 1
    plt.annotate(text=row['abbrev'],
                 xy=(x,y),
                 horizontalalignment='center')

plt.show()
```



Let's add a river. The original data is in a file named 'North_America_Lakes_and_Rivers.zip'. I have filtered the data to retain only the rivers in the Pacific Northwest.


```
In [2... fn = 'nw_rivers.shp.zip'
nw_rivers = gpd.read_file(dbpath + fn)
```

It's important to match the coordinate representation system (CRS) for `nw_states` and `nw_rivers`. The code shows they are already matched, so the `to_crs` call isn't really necessary.

```
In [2... print(nw_states.crs)
print(nw_rivers.crs)
mycrs = nw_rivers.crs
nw_states.to_crs(mycrs, inplace=True)
```

EPSG:4269

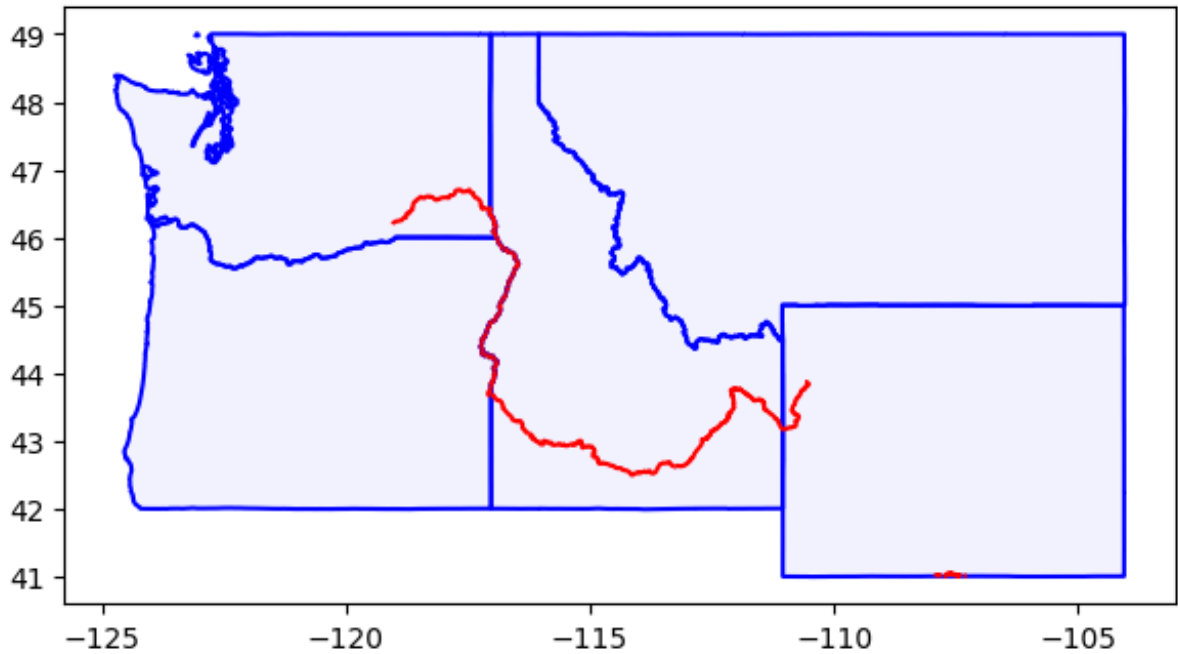
EPSG:4269

```
In [2... def sel(s):
    return nw_rivers['NameEn'].str.contains(s)

Snake = nw_rivers[sel('Snake')]

# remove some small waterways
Snake = Snake[Snake['LengthKm'] > 200]

ax = nw_states.boundary.plot(
    figsize=(7,7), color='blue')
nw_states.plot(ax=ax, color='b', alpha=0.05)
Snake.plot(ax=ax, color='red')
plt.savefig('figs/snake.png', dpi=300)
```



There is one more item to be cleaned up --- the small object at the bottom of Wyoming.

```
In [2... Snake['NameEn']
```

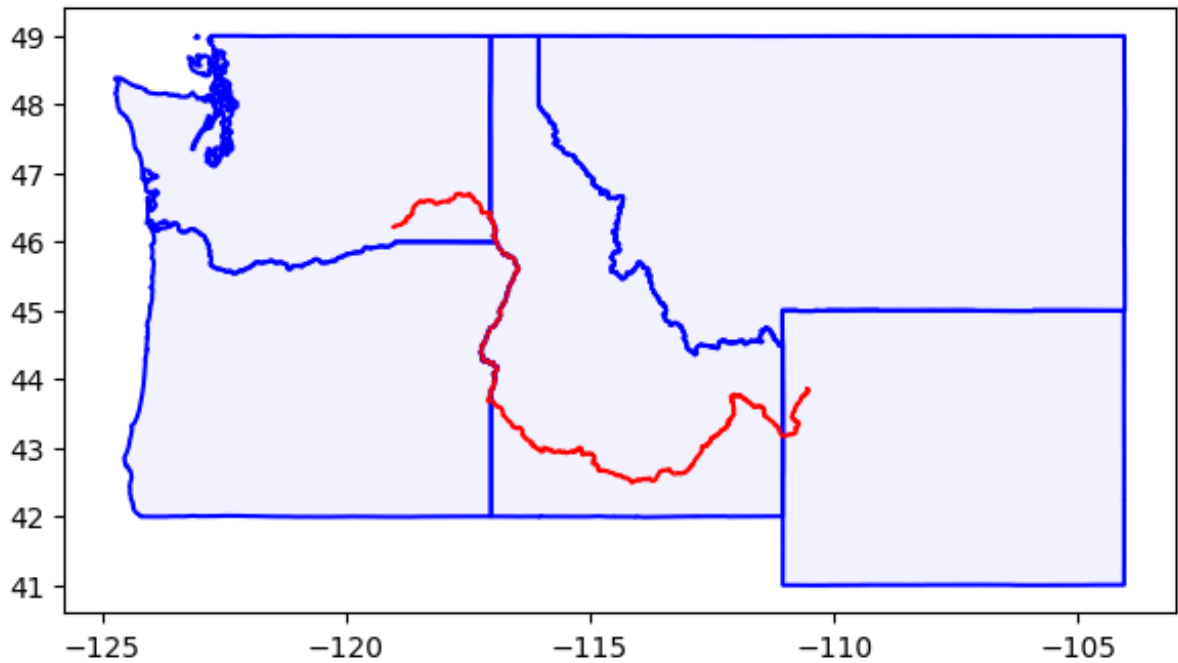
```
Out[2... 57      Little Snake River
112          Snake River
113          Snake River
114          Snake River
115          Snake River
Name: NameEn, dtype: object
```

```
In [2... t = 'Little Snake River'
Snake = Snake[Snake['NameEn'] != t]
```

```
In [2... Snake['NameEn']
```

```
Out[2... 112      Snake River
113      Snake River
114      Snake River
115      Snake River
Name: NameEn, dtype: object
```

```
In [2... ax = nw_states.boundary.plot(
        figsize=(7,7),color='blue')
nw_states.plot(ax=ax,color='b',alpha=0.05)
Snake.plot(ax=ax,color='red')
plt.show()
```



Note: the original rivers data is quite a large file. I filtered it for data in a restricted geographic region and then saved it as a shapefile.

```
In [2... dbpath = '/Users/telliott/data/'
fn = 'North_America_Lakes_and_Rivers.zip'

na_rivers = gpd.read_file(dbpath + fn)
na_rivers = na_rivers.to_crs(mycrs)
nw_rivers = na_rivers.overlay(nw_states,
                              how='intersection')

outpath = '/Users/telliott/Desktop/'
nw_rivers.to_file(
    filename=outpath+'nw_rivers.shp.zip',
    driver='ESRI Shapefile')
```

```
In [2... na_rivers.shape
```

```
Out[2... (5811, 7)
```

```
In [3... nw_rivers.shape
```

```
Out[3... (492, 14)
```

```
In [3... nw_rivers.columns
```

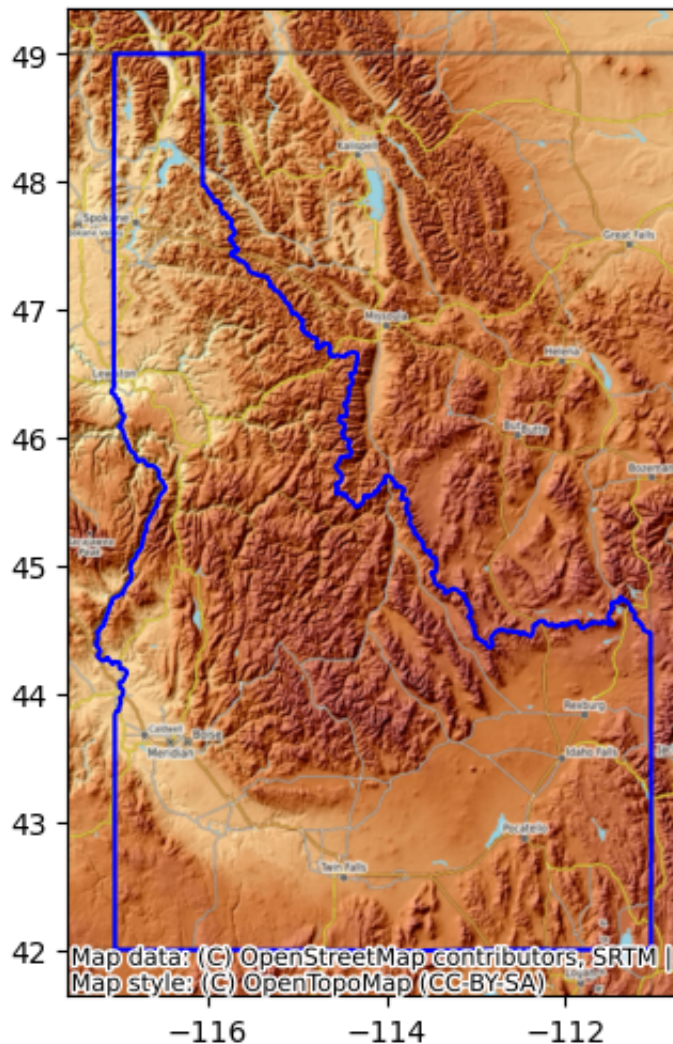
I've picked up some extra columns. I think this happened because I included rivers in British Columbia and Alberta, although we aren't plotting them.

The map displays the Pacific Northwest region, focusing on the Columbia River and its tributaries. The Columbia River is shown in red, flowing from the north and east towards the south and west. The Continental Divide is marked in red. Other rivers shown include the Flathead (black), Missouri (blue), Clark Fork (pink), Snake (pink), Clearwater (blue), Bitter-root (black), Beaver-head (pink), and Yellowstone (red). The locations of Pend Oreille, Clark Fork, Bitter-root, and Flathead are indicated with arrows. The map is bounded by latitude 40 to 52 and longitude -130 to -100.

```
In [3... import geopandas as gpd
import matplotlib.pyplot as plt
import contextily as cx

dbpath = '/Users/telliott/data/'
fn = 'OR_WA_ID_MT_WY.shp.zip'
gdf = gpd.read_file(dbpath+fn)
ID = gdf[gdf['NAME'] == 'Idaho']
ax=ID.boundary.plot(color='b',figsize=(6,6))
```

```
cx.add_basemap(ax, source=cx.providers.OpenTopoMap, crs=ID
plt.show())
```



You have to match the CRS or change (warp) it for the basemap.

clip and cx

The `clip` method "clip[s] points, lines, or polygon geometries to the mask extent." The mask can be a GeoDataFrame, a GeoSeries or a MultiPolygon. If you call clip with a mask like (minx, miny, maxx, maxy), it will call `clip_by_rect`.

The code below illustrates the use of a bounding box and the `.cx` method of a gdf. I've followed the convention for contextily and imported the name as `cx`, even though this could be confusing. Since the method is qualified, Python keeps it all straight.

`cx` may be called like `.cx[xmin:xmax, ymin:ymax]` ; it can also be used with indexes as in `df.cx[:, 1:]`

```
In [3... ax=gdf.boundary.plot()

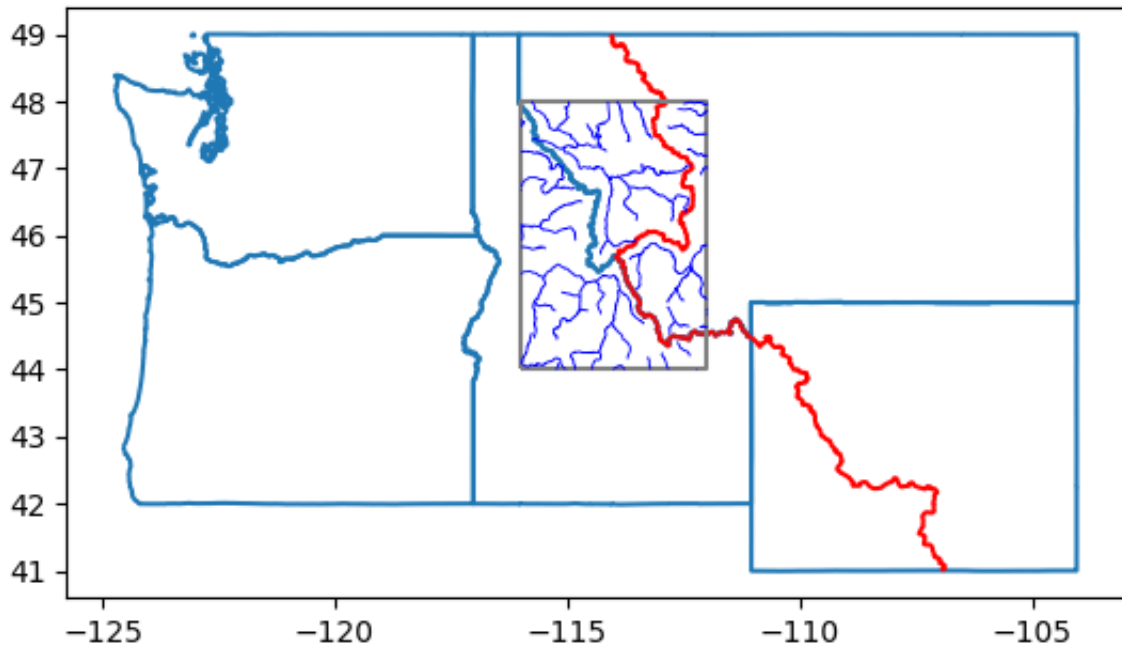
fn = 'nw_rivers.shp.zip'
nw_rivers = gpd.read_file(dbpath + fn)
nw_rivers = nw_rivers.to_crs('EPSG:4269')

# restrict the rivers to a bounding box
xmin, ymin, xmax, ymax = -116, 44, -112, 48
from shapely.geometry import Polygon
poly = Polygon([(xmin,ymin),(xmax,ymin),(xmax,ymax),(xmin,ymax)])
gs = gpd.GeoSeries(poly)
bbox = gpd.GeoDataFrame({'geometry': gs})
bbox = bbox.set_crs('EPSG:4269')
sub = nw_rivers.overlay(bbox, how='intersection')

# continental divide
fn = 'Continental_Divide-Pacific_Atlantic.zip'
cdiv = gpd.read_file(dbpath + fn)
cdiv = cdiv.to_crs('EPSG:4269')
cdiv = cdiv.overlay(gdf, how='intersection')

cdiv.plot(ax=ax,color='r')
bbox.boundary.plot(ax=ax,color='gray')
sub.plot(ax=ax,color='b',lw=0.6)

plt.show()
```



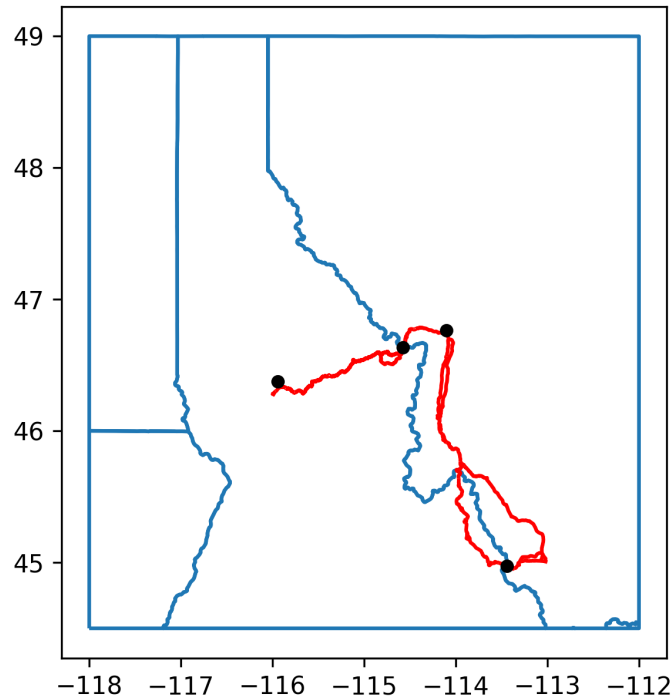
This one is kind of ugly but it illustrates the `bbox` method. I don't know why, but most geometry objects have the LineStrings, Polygons and so on in random order. So you can't just take a slice if you want part of the data.

The red line is the continental divide. This makes it easier to understand which way the rivers must flow.

Lewis and Clark

NPS personnel have accumulated waypoints for Lewis and Clark's travels. I could not actually find the data available from them (although some pages talk about it), but came across a file on ArcGIS (see notes), which was a zipped file (`.kmz.zip`). After unzipping, an online file converter extracted from the `.kmz` a file, `doc.kml` (Keyhole Markup Language) file, that `geopandas` can read with `driver='LIBKML'`.

I was most interested in the part of the trail between the Jefferson River and the Clearwater River, across the continental divide and the Bitterroot Mountains. I added some points: Lemhi Pass, Lolo MT, Lolo Pass, and Weippe ID.



The Lewis and Clark data consist of `LineString` and `MultiLineString` objects. Dealing with these can be a struggle. For example, the components are not in geospatial order.

See [here](#) for how I extracted a "representative point" for each array of coordinates. It shouldn't be that complicated, but it is.

Additional explorations

There is more here than is recorded in this Notebook.

As mentioned, I obtained data for Lewis & Clark's *Corps of Discovery* expedition and plotted some maps from that.

I noticed that there was missing data in the region where the Pend Oreille River flows out of Lake Pend Oreille. I was able to:

- figure out how to use `shapely` to find the two closest points in those features
- confirm the data is entirely missing from the rivers database

- use Google Maps to obtain a few datapoints to approximate the course of the river
- figure out how to use `Geopandas` to make a new `GeoDataFrame` with the new data
- project the extra river points to the right CRS to calculate the length of the `LineString`

All of that is in the directory `Pend Oreille`.

I also wrote up basic cheatsheets for Pandas (in `pandas`) and shapely.

I tried to learn how to use the Overpass GUI to get at OpenStreetMaps data. However that seems beyond me at the moment. Most queries I tried had errors that I couldn't figure out.

And finally, I was able to find a specific feature (Flathead Lake) by using the QGIS application with an OSM plugin (see `notes/osm data.txt`).

Notes about the Jupyter notebook.

To convert this notebook to html do:

```
jupyter nbconvert --execute --to html explore.ipynb
```

One can print directly to a pdf, but I found the font is too big. (This may be because I have the font set large in Safari so I can read it easily).

Saving to html and then printing that seems to work OK. In the github repo, the html looks weird, so at first, I moved it to a subdirectory, but that became awkward.

As the notebook got larger, scrolling to a cell in the middle became impossible. From the web, the fix was

"Settings > Settings Editor > Notebook > Windowing mode > none"

which solved the problem.

Another issue I ran into was that sometimes a Notebook would change such that the menu bar File/Edit ... was obscured by the window below it.

This is a zoom problem, caused by a decrease in the zoom. Sometimes CMD+ will fix it, and sometimes not. It will not be reset by rebooting the notebook.

The solution I found is to go to the real Safari menu bar and do Safari > settings for localhost ... > and then reset the zoom to 100%.